3 Geometry

What students need to learn:

Ref	Content descriptor	Concepts and skills
GM a	Recall and use properties of angles at a point, angles on a straight line (including right angles), perpendicular lines, and opposite angles at a vertex	• Recall and use properties of: – angles at a point – angles at a point on a straight line, including right angles – perpendicular lines – vertically opposite angles • Find the size of missing angles at a point or at a point on a straight line • Distinguish between acute, obtuse, reflex and right angles • Name angles • Estimate sizes of angles • Give reasons for calculations • Use geometric language appropriately • Use letters to identify points, lines and angles • Use two letter notation for a line and three letter notation for an angle • Mark perpendicular lines on a diagram • Identify a line perpendicular to a given line

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GM b	Understand and use the angle properties of parallel and intersecting lines, triangles and quadrilaterals	• Understand and use the angle properties of parallel lines • Mark parallel lines on a diagram • Find missing angles using properties of corresponding and alternate angles • Understand and use the angle properties of quadrilaterals • Give reasons for angle calculations • Use the fact that angle sum of a quadrilateral is 360° • Understand the proof that the angle sum of a triangle is 180° • Find a missing angle in a triangle, using the angle sum of a triangle is 180° • Understand a proof that the exterior angle of a triangle is equal to the sum of the interior angles at the other two vertices • Distinguish between scalene, equilateral, isosceles and right-angled triangles • Understand and use the angle properties of triangles • Understand and use the angle properties of intersecting lines • Use the side/angle properties of isosceles and equilateral triangles

Foundation

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GM c	Calculate and use the sums of the interior and exterior angles of polygons	• Calculate and use the sums of the interior angles of polygons • Use geometrical language appropriately and recognise and name pentagons, hexagons, heptagons, octagons and decagons • Use the sum of angles in irregular polygons • Calculate and use the angles of regular polygons • Use the sum of the interior angles of an n-sided polygon • Use the sum of the exterior angles of any polygon is 360° • Use the fact that the sum of the interior angle and exterior angle is 180° • Understand tessellations of regular and irregular polygons • Tessellate combinations of polygons • Explain why some shapes tessellate and why other shapes do not
GM d	Recall the properties and definitions of special types of quadrilateral, including square, rectangle, parallelogram, trapezium, kite and rhombus	• Recall the properties and definitions of special types of quadrilaterals, including symmetry properties • List the properties of each, or identify (name) a given shape • Draw sketches of shapes • Name all quadrilaterals that have a specific property • Identify quadrilaterals from everyday usage • Classify quadrilaterals by their geometric properties

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GM e	Recognise reflection and rotation symmetry of 2-D shapes	• Recognise reflection symmetry of 2-D shapes • Identify and draw lines of symmetry on a shape • Recognise rotation symmetry of 2-D shapes • Identify the order of rotational symmetry of a 2-D shape • Draw or complete diagrams with a given number of lines of symmetry • State the line symmetry as a simple algebraic equation • Draw or complete diagrams with a given order of rotational symmetry
GM f	Understand congruence and similarity	• Understand congruence • Identify shapes which are congruent • Understand similarity • Identify shapes which are similar, including all circles or all regular polygons with equal number of sides • Recognise that all corresponding angles in similar shapes are equal in size when the corresponding lengths of sides are not equal in size
GM g	Use Pythagoras' theorem in 2-D	• Understand, recall and use Pythagoras' theorem in 2-D
GM i	Distinguish between centre, radius, chord, diameter, circumference, tangent, arc, sector and segment	• Recall the definition of a circle and identify (name) and draw parts of a circle • Understand related terms of a circle • Draw a circle given the radius or diameter

Foundation

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GM k	Use 2-D representations of 3-D shapes	• Identify and name common solids: cube, cuboid, cylinder, prism, pyramid, sphere and cone • Know the terms face, edge and vertex • Use 2-D representations of 3-D shapes • Use isometric grids • Draw nets and show how they fold to make a 3-D solid • Understand and draw front and side elevations and plans of shapes made from simple solids • Given the front and side elevations and the plan of a solid, draw a sketch of the 3-D solid
GM l	Describe and transform 2-D shapes using single or combined rotations, reflections, translations, or enlargements by a positive scale factor and distinguish properties that are preserved under particular transformations	• Describe and transform 2-D shapes using single rotations • Understand that rotations are specified by a centre and an (anticlockwise) angle • Find the centre of rotation • Rotate a shape about the origin, or any other point • Describe and transform 2-D shapes using single reflections • Understand that reflections are specified by a mirror line • Identify the equation of a line of symmetry • Describe and transform 2-D shapes using single translations • Understand that translations are specified by a distance and direction, (using a vector) • Translate a given shape by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$

Ref	Content descriptor	Concepts and skills
GM l	(*Continued*)	• Describe and transform 2-D shapes using enlargements by a positive scale factor • Understand that an enlargement is specified by a centre and a scale factor • Scale a shape on a grid (without a centre specified) • Draw an enlargement • Enlarge a given shape using (0, 0) as the centre of enlargement • Enlarge shapes with a centre other than (0, 0) • Find the centre of enlargement • Describe and transform 2-D shapes using combined rotations, reflections, translations, or enlargements • Distinguish properties that are preserved under particular transformations • Recognise that enlargements preserve angle but not length • Identify the scale factor of an enlargement of a shape as the ratio of the lengths of two corresponding sides • Understand that distances and angles are preserved under rotations, reflections and translations, so that any figure is congruent under any of these transformations • Describe a transformation

Foundation

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GM v	Use straight edge and a pair of compasses to carry out constructions	• Use straight edge and a pair of compasses to do standard constructions • Construct a triangle • Construct an equilateral triangle • Understand, from the experience of constructing them, that triangles satisfying SSS, SAS, ASA and RHS are unique, but SSA triangles are not • Construct the perpendicular bisector of a given line • Construct the perpendicular from a point to a line • Construct the bisector of a given angle • Construct angles of 60°, 90°, 30°, 45° • Draw parallel lines • Draw circles and arcs to a given radius • Construct a regular hexagon inside a circle • Construct diagrams of everyday 2-D situations involving rectangles, triangles, perpendicular and parallel lines • Draw and construct diagrams from given instructions
GM w	Construct loci	• Construct: – a region bounded by a circle and an intersecting line – a given distance from a point and a given distance from a line – equal distances from two points or two line segments – regions which may be defined by 'nearer to' or 'greater than' • Find and describe regions satisfying a combination of loci (NB: All loci restricted to two dimensions only)

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GM x	Calculate perimeters and areas of shapes made from triangles and rectangles	• Measure shapes to find perimeters and areas • Find the perimeter of rectangles and triangles • Find the perimeter of compound shapes • Find the area of a rectangle and triangle • Recall and use the formulae for the area of a triangle, rectangle and a parallelogram • Calculate areas of compound shapes made from triangles and rectangles • Find the area of a trapezium • Find the area of a parallelogram • Find surface area using rectangles and triangles • Find the surface area of a prism
GM z	Find circumferences and areas of circles	• Find circumferences of circles and areas enclosed by circles • Recall and use the formulae for the circumference of a circle and the area enclosed by a circle • Use $\pi \approx 3.142$ or use the π button on a calculator • Find the perimeters and areas of semicircles and quarter circles • Find the surface area of a cylinder
GM aa	Calculate volumes of right prisms and shapes made from cubes and cuboids	• Find the volume of a prism, including a triangular prism, cube and cuboid • Calculate volumes of right prisms and shapes made from cubes and cuboids • Recall and use the formula for the volume of a cuboid • Find the volume of a cylinder